structure of DNA, its replication, the process of making RNA from DNA (transcription), the genetic code that determines the sequence of amino acid in protein synthesis (translation), regulation of gene expression and the essentials of human genome sequencing.

5.1 Gene as the functional unit of inheritance

A gene is a basic physical and functional unit of heredity. The concept of the gene was first explained by **Gregor Mendel in 1860's**. He never used the term 'gene'. He called it 'factor'. In 1909, the Danish biologist Wilhelm Johannsen, coined the term 'gene', that was referred to discrete determiners of inherited characteristics.

According to the classical concept of gene introduced by Sutton in 1902, genes have been defined as discrete particles that follow Mendelian rules of inheritance, occupy a definite locus in the chromosome and are responsible for the expression of specific phenotypic character. They show the following properties:

- Number of genes in each organism is more than the number of chromosomes; hence several genes are located on the same chromosome.
- The genes are arranged in a single linear order like beads on a string.
- Each gene occupies a specific position called locus.
- Genes may exist in several alternate forms called alleles.
- Genes may undergo sudden change in positions and composition called mutations.
- Genes are capable of self-duplication producing their own copies.

One gene-one enzyme hypothesis

The experiments of **George Beadle and Edward Tatum** in the early 1940's on *Neurospora crassa* (the red bread mould) led them to propose one gene-one enzyme hypothesis, which states that one gene controls the production of one enzyme.

One gene-one polypeptide hypothesis

It was observed that an enzyme may be composed of more than one polypeptide chain and a gene can code for only one polypeptide chain. Thus one gene-one polypeptide hypothesis states that one gene controls the production of only one polypeptide chain of an enzyme molecule.

5.2 In search of the genetic material

As early as 1848, Wilhelm Hofmeister, a German botanist, had observed that cell nuclei organize themselves into small, rod like bodies during mitosis called **chromosomes**. In 1869, Friedrich Miescher, a Swiss physician, isolated a substance from the cell nuclei and called it as **nuclein**. It was renamed as nucleic acid by Altman (1889), and is now known as DNA. By 1920, it became clear that chromosomes are made up of proteins and DNA. Many experiments were carried out to study the actual carriers of genetic information. Griffith's experiment proved that DNA is the genetic material which has been dealt in class XI. Bacterial transformation experiments provided the first proof that DNA is the genetic material. However, he could not understand the cause of bacterial transformation, and the biochemical nature of genetic material was not defined from his experiments.

Later, Oswald Avery, Colin Macleod and Maclyn McCarty in 1944 repeated Griffith's experiments in an '*in vitro*' system in order

XII Std Zoology Chapter 5 EM.indd 62 23/12/2022 16:07:09